

Graphing Quadratics Test

Advanced Algebra 2 (10th-11th Grade)

Instructions: Show all work for full credit. Write answers clearly in the spaces provided.

Total Points: 100 Time: 45 minutes

Part 1: Depth of Knowledge Level 1 - Recall & Reproduction (15 points)

1. **(3 points)** For the equation $y = -3(x + 5)^2 - 7$, identify the vertex.
2. **(3 points)** What is the axis of symmetry for $y = (x - 4)^2 + 2$?
3. **(3 points)** Does the parabola $y = -\frac{1}{2}(x - 1)^2 + 3$ open up or down?
4. **(3 points)** Complete the square: $y = x^2 + 10x + \underline{\hspace{2cm}}$ becomes a perfect square trinomial.
5. **(3 points)** For $y = 4(x - 2)^2 + 1$, is the parabola wider or narrower than $y = x^2$?

Part 2: Depth of Knowledge Level 2 - Skills & Concepts (30 points)

6. **(8 points)** Convert $y = x^2 + 8x + 7$ to vertex form by completing the square. Show all steps.

Vertex form: _____

Vertex: _____

7. **(8 points)** Convert $y = 2x^2 - 12x + 10$ to vertex form. Show all work.

Vertex form: _____

8. **(7 points)** Describe all transformations of $y = -2(x + 3)^2 - 5$ compared to the parent function $y = x^2$.
9. **(7 points)** Given the vertex $(2, -3)$ and knowing the parabola opens upward with a vertical stretch by a factor of 3, write the equation in vertex form.

Part 3: Depth of Knowledge Level 3 - Strategic Thinking (30 points)

10. **(10 points)** A parabola has its vertex at $(-2, 5)$ and passes through the point $(0, 1)$.
- (a) Find the value of a in the vertex form equation.
 - (b) Write the complete equation in vertex form.
11. **(10 points)** Given $y = x^2 - 6x + 11$:
- (a) Convert to vertex form.
 - (b) Identify the vertex.
 - (c) Explain why this parabola has no x-intercepts (zeros).
12. **(10 points)** A student converted $y = 2x^2 + 12x + 10$ to vertex form and got $y = 2(x + 3)^2 - 8$.
- (a) Verify if this is correct by expanding their answer.
 - (b) If incorrect, find and explain the error, then provide the correct vertex form.

Bonus Question (5 points extra credit)

14. **(5 points)** A parabola in vertex form is $y = a(x - h)^2 + k$. Explain how you could determine the value of a if you are given the vertex (h, k) and one other point (x_1, y_1) on the parabola. Then demonstrate your method using vertex $(3, -4)$ and point $(5, 4)$.

END OF TEST

Good luck!